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GEOG441

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13 April 2026

Using Sentinel-2 Data to Analyze Mangrove Fragmentation and Recovery Time Following Hurricane Maria in Puerto Rico

Introduction

On the morning of September 20th, 2017, Hurricane Maria made landfall on the southeast coast of Puerto Rico. Striking the island as a category 4 storm, peak recorded wind speeds at this time were around 135kt, however, at some elevations winds are expected to have been above 140kt (Pasch et al., 2018). In the 8 hours that the storm moved over Puerto Rico, some locations saw a maximum daily rainfall of over 800mm (Keellings, 2019). Monitoring stations at high elevations in Caguas and Villalba saw between 685-935mm of recorded rainfall (Pasch et al., 2018). This is about a fourth of the wettest point on the island's average annual rainfall of 3500-4000mm/yr (Colon-Torres, 2009). Out of 129 storms that have made landfall or impacted Puerto Rico since 1956, Maria had the largest maximum daily rainfall and maximum mean total rainfall at monitoring stations across varying elevations (Keellings, 2019). The storm departed the island into the western Atlantic with wind speeds of 95kt, just below category 3 intensity. The combined surge and tide produced by the storm reached maximum inundations of 6-9 feet above ground level along the Ceiba municipalities. On the southeast coast, water levels were measured as high as 2.8 ft, with waves contributing to another 2 feet to the water level (Pasch et al., 2018).

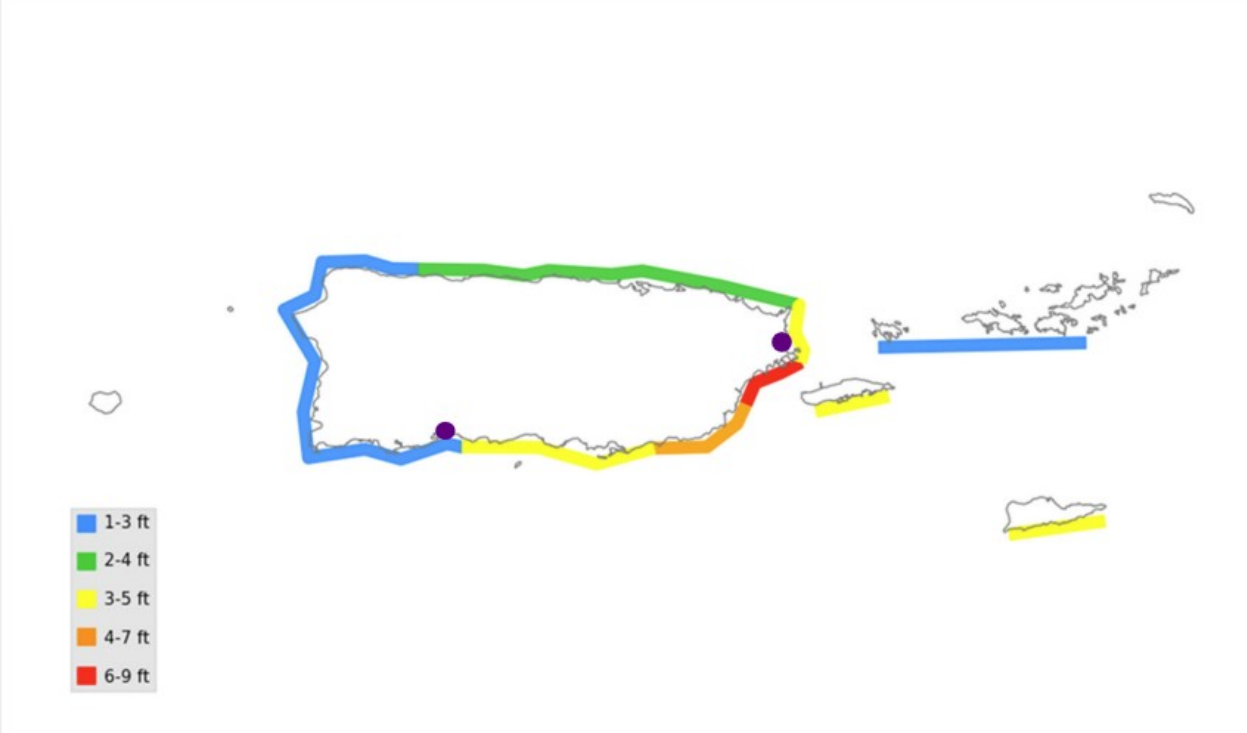


Figure 1. Credit NHC Storm Surge Unit: Shows estimated maximum storm surges across island of Puerto Rico. Purple marker on southern coast shows Puerto de Guayanilla. Purple marker on eastern coast shows Ceiba municipality.

On average, households went 84 days without electricity, 68 days without water, and 41 days without cell service from September 20th until December 31st, 2017—with conditions worsening across more remote communities (Kishore et al., 2018). Maria sits as the fourth costliest hurricane in U.S. history, with total damages in Puerto Rico and the U.S. Virgin Islands summing to an estimated 115.2 billion dollars (National Oceanic and Atmospheric Administration). While the official death count is 64, this number is widely disputed, most famously by the mortality paper published by Kishore et al. (2018) which estimates an additional 4645 deaths attributed to Hurricane Maria. The study found the total excess deaths between

directly related (drowning, struck by debris, etc.) and deaths indirectly related (loss of services, loss of medical care, etc.) to be 70 times the official amount.

Coastal wetlands provide protection towards local communities from hurricanes and tropical storms (Costanza et al., 2008). Specifically, mangrove ecosystems provide storm protection value by reducing water height and velocity of storm surges. Mangroves are a group of plants that grow in sub-tropical latitudes, existing in areas of high temperatures, extreme tides, and strong winds (Kathiresan and Bingham, 2001). They provide physical drag against wave energy with their trunks, leaves, and roots. The root systems of mangroves increase sediment accretion which works against coastal erosion (Marois and Mitsch, 2015). The urban invasion of anthropogenic threats towards mangrove habitat in the Caribbean has been mapped over the last 20 years. While studies have shown the negative effects of anthropogenic activities on mangrove health, there is not enough evidence to suggest that urbanization plays a large role in tropical storm recovery in Puerto Rico's mangroves (Branoff, 2020). Instead, mortality and recovery periods were found to vary across geomorphology, species, and size (Branoff, 2020). While some studies have found that mangrove loss is often associated with high levels of fragmentation (Bryan-Brown, 2020), there is little research on mangrove fragmentation and tropical storm recovery period.

This study aims to provide insight on the response of Puerto Rico's mangrove forests between fragmented and non-fragmented areas of wetland ecosystems. The conservation of coastal forest zones has been debated since the mid-19th century. While the importance of mangroves and the threats to their persistence have been recognized globally, the cooperation and effective solutions for conservation have differed across regions of the world (Romanach et al., 2018). Increased research and usage of remote sensing techniques through modern missions

like Landsat-9, Sentinel-2, and TERRA, in tandem with enhanced field-work methods will further optimize conservation efforts.

Discussion

Study Area

Guayanilla Bay, or *Puerto de Guayanilla* is a semi-enclosed shallow bay located on the southern coast of Puerto Rico (18°N; -66.76°W). A narrow tidal inlet is formed by a hook-shaped peninsula stretching from the southeast, enclosing the lagoon from the Caribbean Sea to the south. The natural harbor is lined with a thin strip of dense mangroves extending no more than 200m inland at its widest point, with short fringe vegetation extending for another 50-100m inland (Fig. 3). When vegetation ends between 150-350m from the coast, land use becomes a mix of crops, rangeland, and grassland. There are no industrial structures, and there is minimal urban development in the immediate study area. The thin clusters of mangrove coverage and lack of inland habitat development make for a spatially fragmented ecosystem (Fig. 3).

Ceiba State Forest, or *Bosque Estatal de Ceiba* is a protected nature reserve located on the northeast coast of Puerto Rico (18.20°N; -65.55°W). The reserve is composed primarily of a large mangrove forest covering over 350 acres (130-180 hectares) which sits on an open coastline facing the Atlantic (Fig. 4). At its widest point the mangrove forest extends 800m inland, with an average width of 500m across the entire 1.5 miles (2.4km) of coastline that the reserve occupies (Fig. 5). The mangrove forest is composed primarily of red mangrove (*Rhizophora mangle*), black mangrove (*Avicennia germinans*), and white mangrove (*Laguncularia racemosa*), providing dense, diverse, and contiguous coverage. Ceiba State Forest

is characterized by exceptionally high structural complexity supporting biomass levels of up to 245 tons per hectare (Weaver and Gould, 2013).



Figure 2. Credit Sources, Esri, Digital Globe, GeoEye, i-cubed, USDA FSA, USGS, AEX, Getmapping, Aerogrid, IGN, IGP, swisstopo, and the GIS User Community: Satellite imagery shows study area of Guayanilla Bay. Dense mangroves shown as dark green vegetation along coast in center of image. Shorter fringe mangrove observed as lighter greens 50-150m inland from coast.

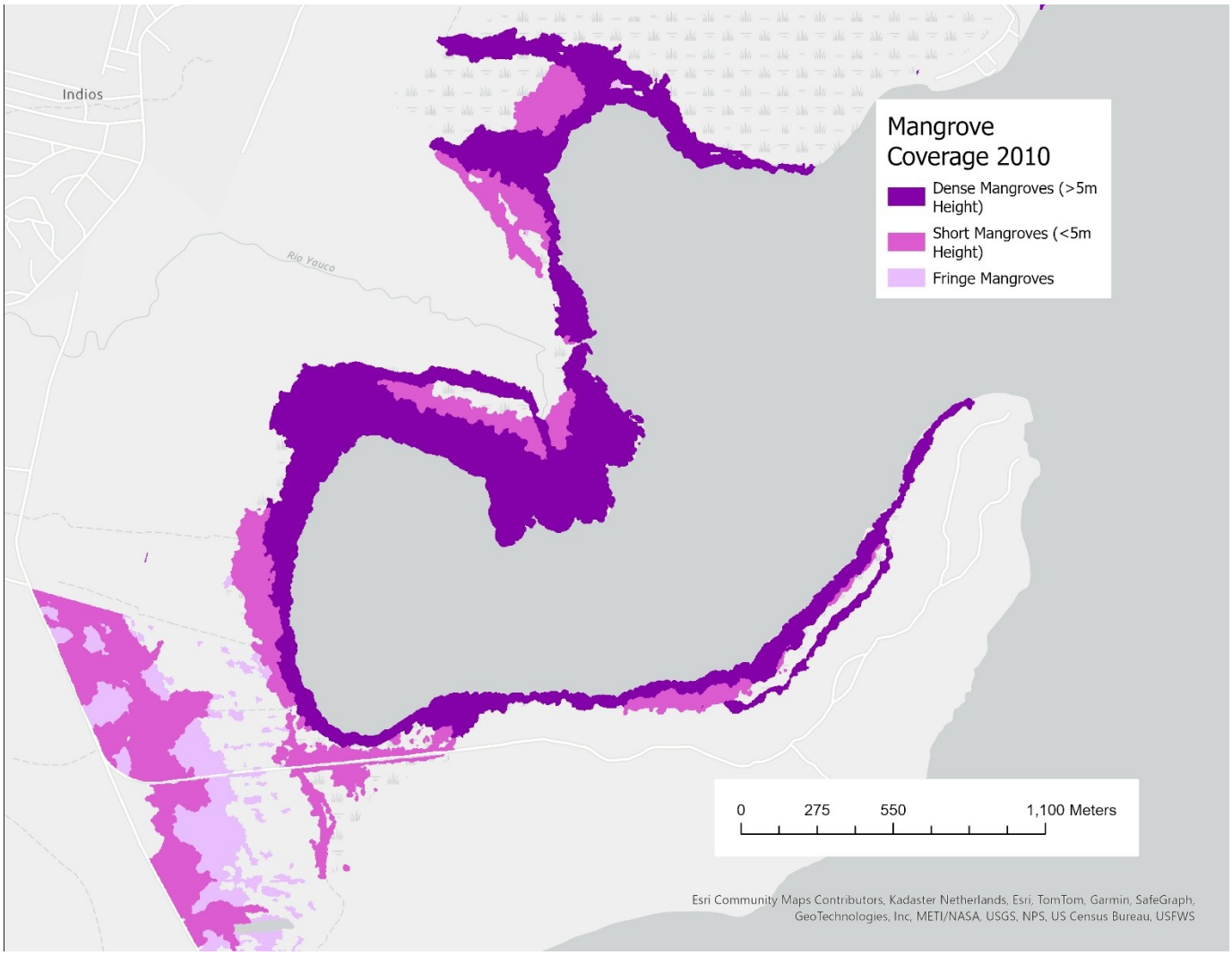

Indios
Rio Yauco
Mangrove Coverage 2010
Dense Mangroves (>5m Height)
Short Mangroves (<5m Height)
Fringe Mangroves
0
275
550
1,100 Meters
Esri Community Maps Contributors, Kadaster Netherlands, Esri, TomTom, Garmin, SafeGraph, GeoTechnologies, Inc, METI/NASA, USGS, NPS, US Census Bureau, USFWS

Figure 3. Credit National Oceanic and Atmospheric Administration, Office for Coastal Management. “2010 NOAA Puerto Rico C-CAP 30m Land Cover.”



Coastal Change Analysis Program (C-CAP) Regional Land Cover: Figure shows mangrove coverage in Guayanilla Bay from 2010. Dense mangrove coverage taller than 5 meters in height shown in dark purple. Short mangroves below 5 meters in height are shown in pink. Fringe/shrub mangroves are shown in light pink.

Figure 4. Credit Sources, Esri, Digital Globe, GeoEye, i-cubed, USDA FSA, USGS, AEX, Getmapping, Aerogrid, IGN, IGP, swisstopo, and the GIS User Community: Satellite imagery shows study area of Ceiba State Forest located in center of image, characterized by dense dark green vegetation. Bordered by barren soil to the west and the Atlantic to the east. Mangroves end 500-1000m inland, where grassland, urban areas, and rangeland continue west.

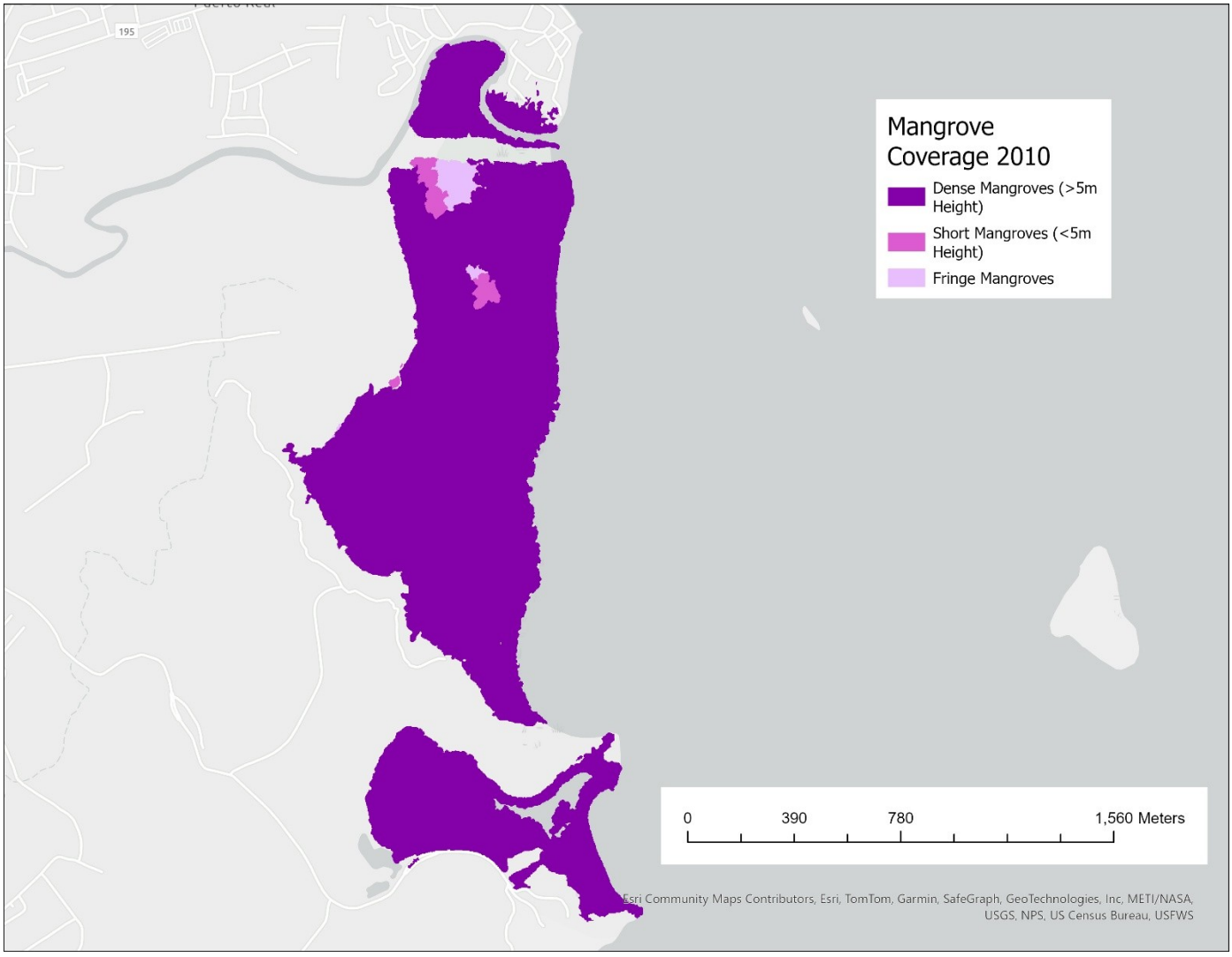


Figure 5. Credit National Oceanic and Atmospheric Administration, Office for Coastal Management. “2010 NOAA Puerto Rico C-CAP 30m Land Cover.” Coastal Change Analysis Program (C-CAP) Regional Land Cover: Figure shows mangrove coverage in Ceiba State Forest. Coverage is almost exclusively dense, high canopy mangroves shown in dark purple. Some fringe/shrub mangroves areas found in the northern regions, indicated by pink.

Methods

To accurately compare mangrove recovery across different regions of affected coastline, I gathered NOAA C-CAP High-Resolution Land Cover data. While post-storm data was not available for Puerto Rico, U.S Virgin Islands, and Guam, high-resolution (10m) data from 2010-2017 provided a baseline for mangrove coverage prior to Maria. The three extracted land cover types used were Estuarine Forested Wetland (tidal wetlands dominated by woody vegetation >5 meters in height), Estuarine Scrub Wetland (tidal wetlands dominated by woody vegetation <5 meters in height), and Estuarine Emergent Wetland (tidal wetlands dominated by rooted hydrophytes where vegetation cover exceeds 80 percent). This data will include the mangrove species *Avicennia germinans*, *Laguncularia racemosa*, *Rhizophora mangle*, and other species which account for differences in recovery between forests, providing an aggregate of coverage data (Branoff, 2020). In addition, ALOS-2 JAXA raster data from the Global Mangrove Watch is included. Coverage includes data from 2007-2020 providing pre-storm and post-storm mangrove coverage displaying immediate and long-term effects. Sentinel-2 data is used to provide vegetation indices and satellite imagery from 2016-2026. EVI2 (Enhanced Vegetation Index 2) is used to monitor change over time, an effective vegetation index enhancing NDVI in areas of

dense canopy cover and mangroves found in the study area. EVI2 provides increased accuracy across range when describing landscape characteristics in discrete land cover classes, making it ideal for tropical application (Mondal, 2010). The BSS (Barren Soil Script) is an enhancement of BSI (Bare Soil Index) with NIR B08 applied to the green channel and SWIR band B11 applied to the red channel (Nguyen, 2021). The script displays all vegetation in green and water in black. Barren ground appears red, suitable for monitoring habitat loss, ecosystem degradation, and recent erosion. It is noted that both indices depend on seasonal variation, so observations are made at the same annual point through the study period.

Findings

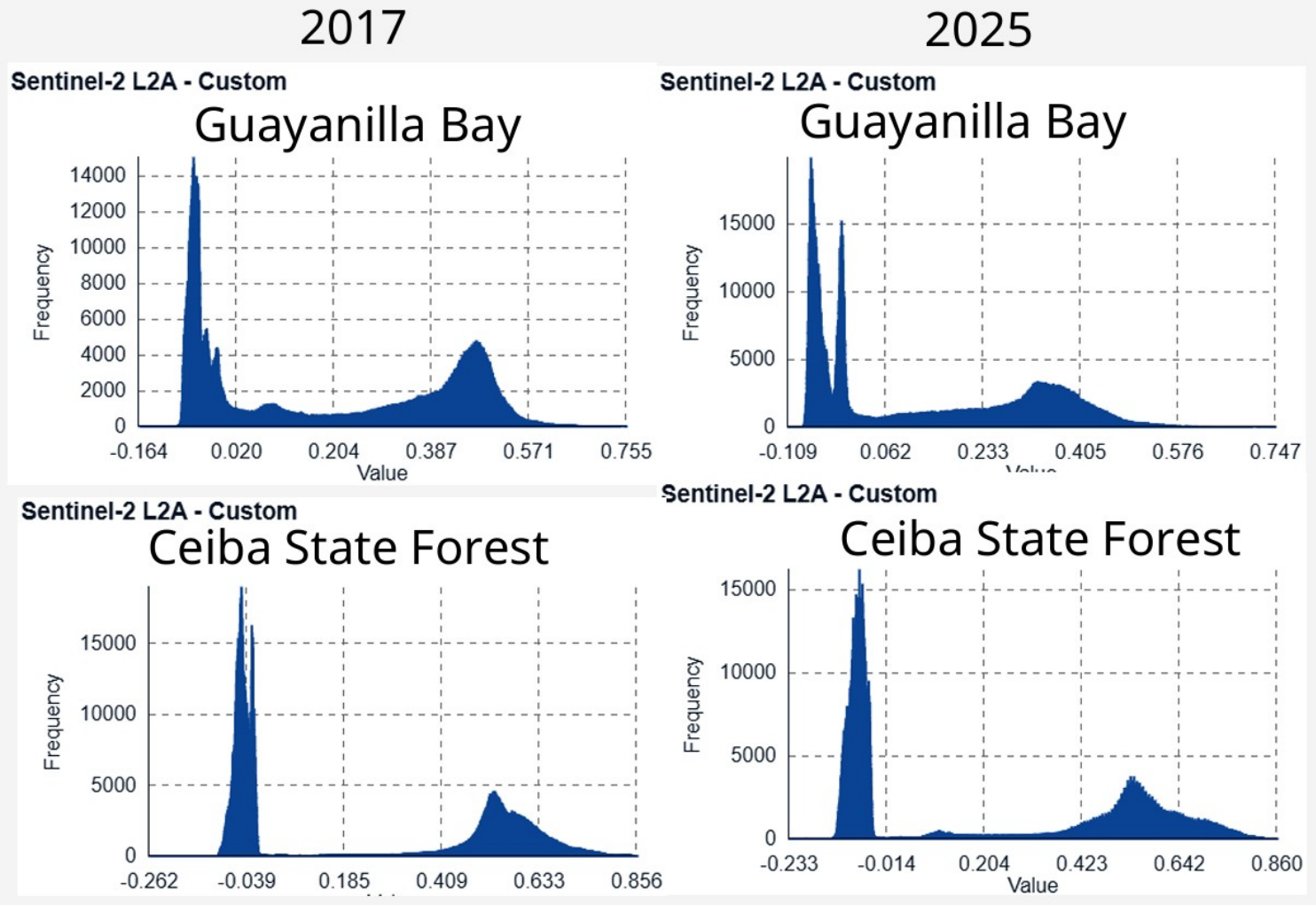


Figure 6. Credit Copernicus Sentinel data 2017-2025. Retrieved from ASF DAAC 4/14/2026. Processed by ESA. Above figure shows charts containing EVI2 data from both Guayanilla Bay and Ceiba State Forest, showing changes in value from 2017-2025.

At Ceiba State Forest, EVI2 data shows vegetation index values returning to pre-storm levels by 2020, with some areas exceeding pre-storm levels by 2025 (Fig. 6). JAXA mangrove raster data and Sentinel-2 imagery confirm the initial loss following the storm followed by the subsequent recovery across the entire reserve. The dense canopy structure, species diversity, and geomorphology provide ideal conditions for seed regeneration and nutrient retention. Despite having faced stronger storm surge, peak wind speeds, and greater rainfall totals, Ceiba saw relatively rapid recovery. At Guayanilla Bay, fragmented mangroves have not experienced comparable recovery (Fig. 6). EVI2 data shows continuous vegetation deterioration annually from 2017-2025, with some regions showing complete loss by early 2022. BSS and Sentinel-2 imagery confirm increasing barren ground exposure along the coastline. Thin fragmented mangrove structures at Guayanilla lack the properties to support quick recovery, limiting nutrient and sediment delivery necessary for regrowth.

Additionally, in 2020 a 5.8 magnitude earthquake struck near Guayanilla, which may have further compounded degradation at the site. However, due to deterioration trends prior to 2020 there is sufficient evidence to suggest that fragmentation and geomorphology are the primary drivers for the lack of resurgence. Ceiba, although having faced more severe storm impact, benefits from having favorable contiguous regeneration properties. Guayanilla Bay’s thin

fragmented mangrove structure lacks interior forest support. This suggests that mangrove spatial configuration is a stronger predictor of recovery period than storm intensity alone.

Conclusion

Moreover, fragmented mangrove forests show significantly slower post-storm recovery times than contiguous forests. Ceiba State Forest recovered to pre-storm vegetation levels within three years of Hurricane Maria, exceeding pre-storm levels by 2025. Meanwhile, Guayanilla Bay continues to deteriorate almost a decade after the storm. Patch size, inland extent, and geomorphology appear to be strong predictors of mangrove resiliency during post-storm periods. Results suggest that conservation efforts should prioritize protection of contiguous mangrove habitat and expand protection in fragmented areas. Contiguous mangrove systems host ideal properties for seed regrowth and regeneration (Woznicki et al., 2022). Further research incorporating additional fragmented and contiguous forest areas across the Caribbean would strengthen these findings and provide more comprehensive insight for coastal management strategies.

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